

PARITY_GAPS

Python → Go Parity Gaps

Revision: 1 **Last modified:** 2026-06-06T00:00:00Z

Last audited: 2026-04-27 (commit 5f12be0) **Audit method:** side-by-side read of public surface (grep -E "^func" on Go, module-level inspection on Python). No behavior testing performed. **Spec reference:** docs/superpowers/specs/2026-04-26-jackett-autoconfig-clean-rebuild-design.md §7

Status definitions

- **Ported** — feature exists in Go with equivalent behavior. Note inline divergences.
- **Partial** — feature exists in Go but lacks a sub-behavior. Note the gap.
- **Missing** — feature has no Go counterpart. Note user-visible risk.

Summary

Status	Count
Ported	6
Partial	4
Missing	8
Total Python features audited	18

Matrix

download-proxy/src/merge_service/

Python module	Feature	Go location	Status	Risk
search.py			Partial	

Python module	Feature	Go location	Status	Request
	start_search() orchestrator	internal/service/ merge_search.go: StartSearch		Go search start process Private
search.py	_classify_plugin_stderr() error categorization	(none)	Missing	N e P p o
search.py	Public tracker subprocess + NDJSON parser	(none)	Missing	
search.py	Private tracker direct aiohttp (RuTracker, Kinozal, NNMCclub, IPTorrents)	Partial in internal/service/ merge_search.go	Partial	V p
deduplicator.py	Tiered dedup (infohash → name+size → fuzzy)	merge_search.go (inlined)	Partial	V P D h t f S p t N r N m p N c s
enricher.py	TMDB title resolution	(none)	Missing	
enricher.py	OMDb fallback	(none)	Missing	S
enricher.py	TVMaze / AniList / MusicBrainz / OpenLibrary	(none)	Missing	
validator.py	BEP 48 HTTP scrape	(none)	Missing	
validator.py	BEP 15 UDP scrape + bencode parser	(none)	Missing	
scheduler.py	Recurring search driver (cron-like)	internal/api/ scheduler_api.go: ScheduleStore	Partial	G s l t s
hooks.py	Pre/post-search + download hooks (callback dispatch)	internal/api/ hooks.go: HookStore	Partial	S r (E r a b G c
retry.py	Shared retry policy via tenacity	(none)	Missing	
jackett_autoconfig.py		(none)	Missing	

Python module	Feature	Go location	Status	R
	Env-discovery + Jackett indexer auto-config (NEW in this spec)			o U

download-proxy/src/api/

Python module	Feature	Go location	Status
routes.py	merge/search REST endpoints	internal/api/search.go	Ported
auth.py	Private-tracker auth (login, session refresh, CAPTCHA)	(none)	Missing
hooks.py	hooks REST endpoints	internal/api/hooks.go	Ported
scheduler.py	scheduler REST endpoints	internal/api/scheduler_api.go	Ported (storage only — see scheduler driver gap above)
streaming.py	SSE streaming endpoint	internal/service/sse_broker.go + internal/api/search.go	Ported
theme_state.py	Dashboard theme state	internal/api/theme.go	Ported
jackett.py	/api/v1/jackett/autoconfig/last (NEW)	(none)	Missing

download-proxy/src/ui/

Python module	Feature	Go location	Status
Jinja2 dashboard templates	Operator UI fallback	(none)	Missing

The Angular SPA dashboard is shared (lives in frontend/, served by both). The Jinja2 fallback dashboard is Python-only.

User-visible regressions if you flip Go-only today

In priority order:

1. **No metadata enrichment.** Search results lose posters, year, content type. Highest user-visible degradation.
2. **No Jackett auto-config.** Every fresh boot requires manual Jackett UI clicks.
3. **No public tracker plugin fan-out.** 600+ Python plugins (eztv, piratebay, etc.) unreachable. Search would only return whatever Jackett aggregates + the 4 private trackers (and only those wired in Go).
4. **No tracker validation.** Stale peer counts, no health detection.
5. **Schedules silently no-op.** UI creates them, driver never fires.
6. **No private-tracker auth REST surface.** No CAPTCHA refresh flow.

Per-gap follow-up specs (proposed, in priority order)

1. **enricher.go** port — biggest user-visible gap. Wraps 6 external APIs. Estimate 2-3 days with tests.
2. **validator.go** port — BEP 48/15 with bencode parser. Estimate 1-2 days.
3. **scheduler.go driver loop** — make existing storage actually fire schedules. Estimate 1 day.
4. **Public tracker plugin subprocess fan-out** — replicate `_search_tracker()` subprocess pattern. Estimate 2-3 days.
5. **jackett_autoconfig.go** port of `jackett_autoconfig.py`. Estimate 1 day.
6. **auth.go** REST surface for private-tracker auth flows. Estimate 1-2 days.
7. **hooks.go runtime dispatch** — verify and complete callback firing.
8. **retry.go** shared policy.
9. **deduplicator.go** explicit tiered structure (clarification, not new behavior).

Each gap should ship as its own spec → plan → implementation cycle in the superpowers flow, so nothing gets lost during the eventual Go-only flip.

Audit data points captured

- Go service exports (count): 18 functions across `merge_search.go` + `sse_broker.go`.
- Go API exports (count): ~30 handler/store functions across 6 files.
- Python `merge_service/` modules: 7 (deduplicator, enricher, hooks, `jackett_autoconfig`, retry, scheduler, search, validator).
- Python `api/` modules: 7 (auth, hooks, `jackett`, routes, scheduler, streaming, `theme_state`).